MACOM
Partners from RF to Light

# CS International GaN/Si for RF Applicati

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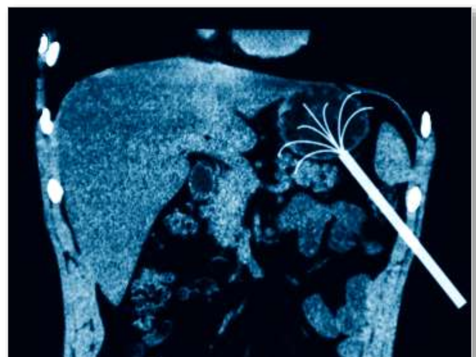
# Shattering the Barriers to Mainstream GaN Adoption

# Outline

- The GaN Revolution
- GaN/ Si Positioning
- GaN Misinformation
- SiC and Si Materials Comparison
- Product Comparisons
- Reliability
- Integration Potential
- Future Directions
- Conclusions





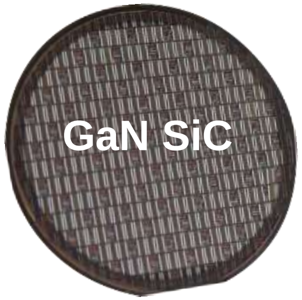
# The GaN Revolution

## MACOM is Driving GaN to Mainstream Commercial Adoption

- **GaN-on-SiC**
  - Small Diameters
  - Small, Boutique FABs
  - Limited Ability to Scale
- **GaN-on-Silicon**
  - Large Wafer Diameters
  - Leverage High Volume Silicon
    - Processing Capability/Discipline
    - Supply Chain Robustness
    - Cost Structure
  - Path to full silicon integration

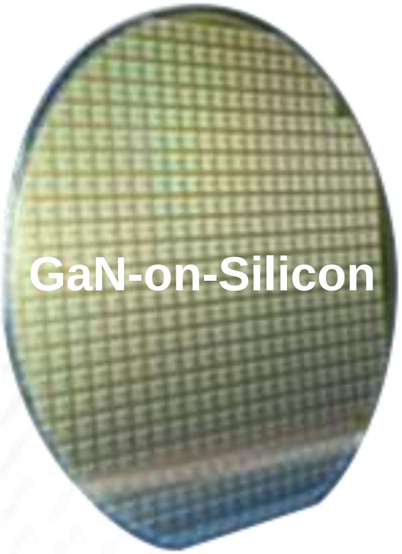
**Takeaway:** GaN Must Compete on BOTH Cost and Performance to Win in the Market

**GaN Performance at LDMOS Prices!!**




**US Defense Market**

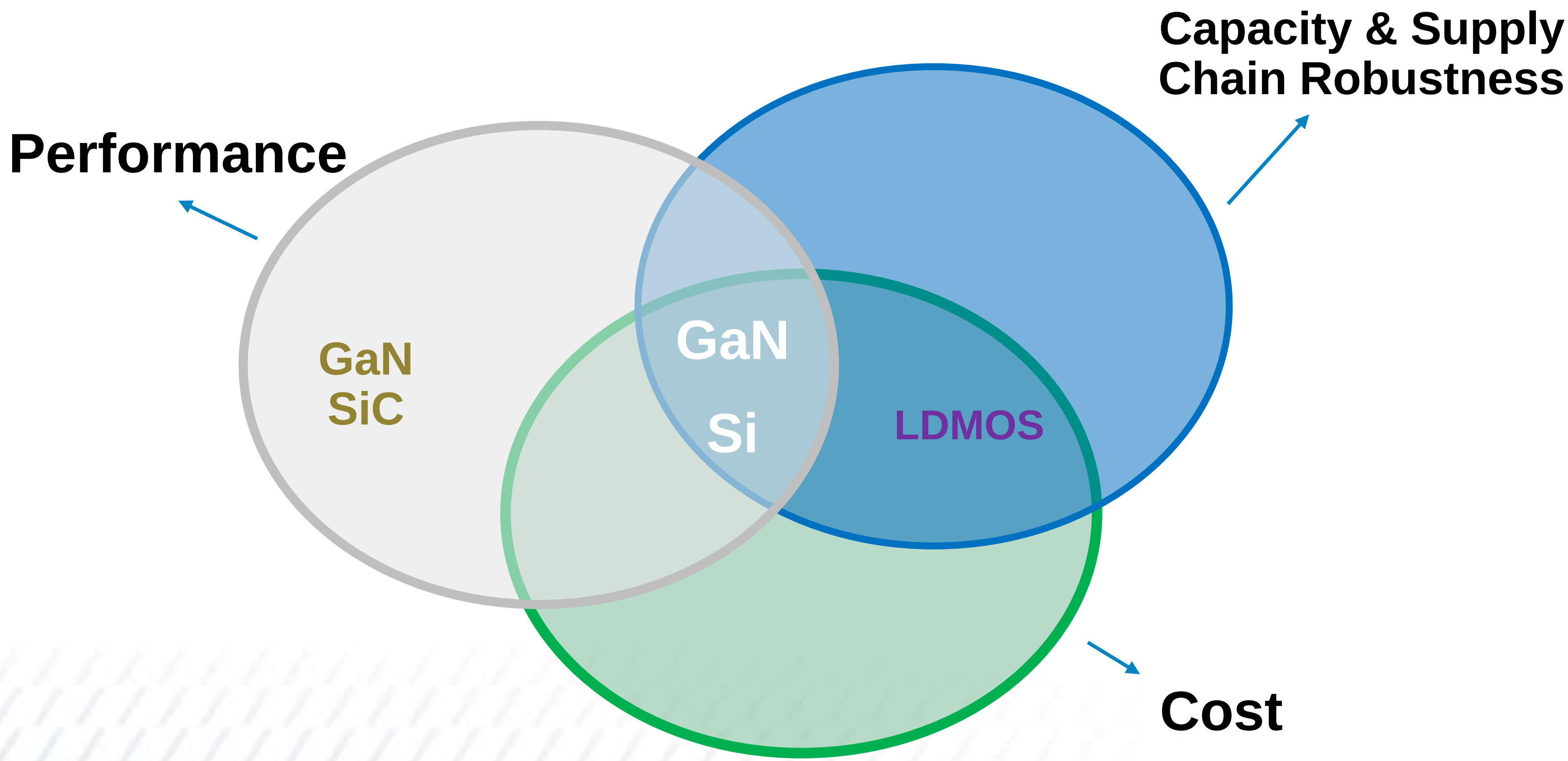
**Small Wafers – Small FABs**




**Global Commercial Market**

**Large Wafers – High Volume FABs**

# GaN/Si Positioning




**GaN/Si offers the best of all three requirements: performance, cost, and scalability and supply chain robustness**

# GaN Misinformation

**FICTION**

***"GaN on Si is limited to just 100W output power."***

***"GaN on Si products have poor thermal conductivity and cannot dissipate heat as efficiently as GaN on SiC"***

***"GaN on Si can not be housed in small, plastic packaging while reliably dissipating heat"***

***"GaN on Si doesn't come close to GaN on SiC performance"***

# The Market Sets the Price

| Market | Typical Device | Price Point |
| --- | --- | --- |
| Aerospace and Defense | Broadband, High Efficiency, MMIC HPA<br>• 10 – 1000 W<br>• 28/48 V operation | \$2.00 – \$5.00/Watt |
| CATV | Low Frequency, Broadband, High Linearity | \$0.5 – \$1.00 |
| Infrastructure | Discrete Power Transistors | \$0.15 – \$0.30/Watt |

- The lowest cost solution ***always*** wins in the marketplace
- So …. If Silicon can do …. Silicon will win ….
- So …. If Silicon can't & GaAs can …. GaAs will win ….
- GaN needs to provide unique solutions & provide compelling value to win

# GaN/Si vs GaN/SiC
# Substrate Materials Technology Drives Cost

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| Device Technology | Substrate Technology | Substrate Maturity | Size | Industry Volume | $/in$^2$ |
|---|---|---|---|---|---|
| GaN on SiC | SI-SiC | Low | 3" - 6" | 1 X | 1 X |
| GaN on Si | Silicon | High | 4" – 12" | $10^6$ X | $10^{-3}$ – $10^{-4}$ X |

## > Cost Structure

- SiC Substrate Material Generation – 150 μm/hr at >2200°C
- SiC Substrate Production Dominated by Single Supplier
- Silicon Substrate Material – 3 in/hr (76,200 μm/hr) at ≈1400°C
- Silicon Substrates – Many Suppliers/Very Competitive Market
- Substrate Cost/Area is Factor of >1000x - Si:SiC
- GaN Epitaxial Growth Differences Immaterial
- Substrate Cost Dominates GaN Epitaxial Material

Silicon Carbide World

## > Effect on Thermal Design

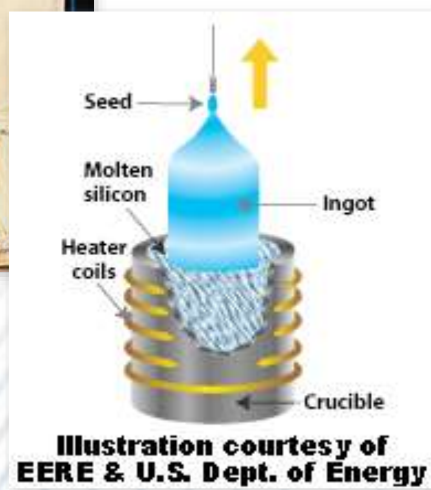
- Improve Device Thermal Resistance on Silicon Substrate
- Thermal Resistance Sets the Reliable Device Operating Junction Temperature
- Spread Heat Generating Areas => Larger Die
- Die Size Increase ≈20%
- Enabled by Lower Cost of Silicon Substrate

Silicon World

# Silicon vs. SiC Wafer Substrate Production

Silicon Ingot

(6” wafer = $20)

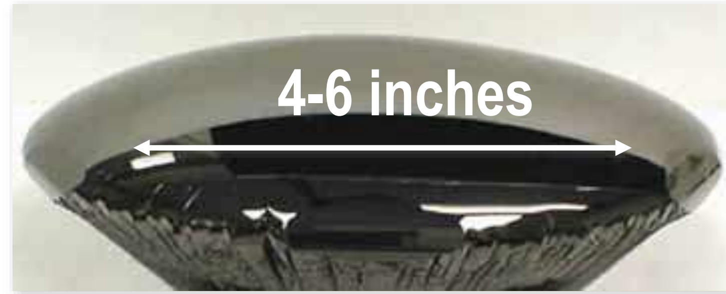



Silicon Ingot Pulling

≈1400° C

>3 inches/hr

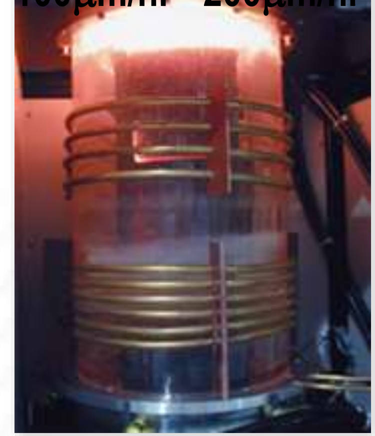
SiC Ingot (6” wafer = $6000)




High Temperature (>2000° C)

PVT SiC Ingot Growth Reactor

100µm/hr - 200µm/hr

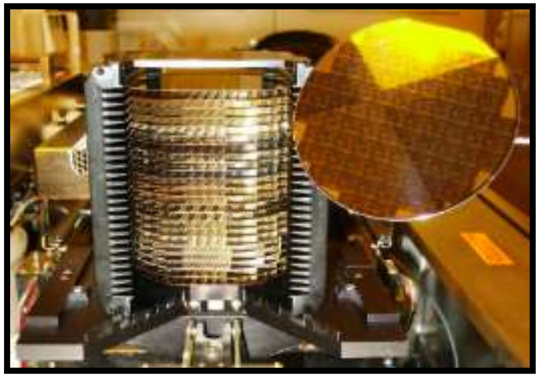


**3000 Si wafers from a single ingot vs. < 100 SiC wafers from a SiC “ingot”**

# GaN/Si RF Device Differentiation is High Quality Silicon Substrates

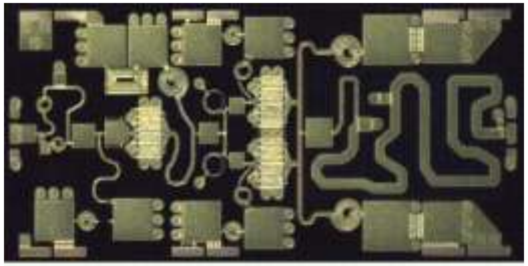
## > Float-zone Silicon Substrates

- No Defects
- Scalable to 200-mm and beyond
- Mature supply chain
- High Resistivity Float Zone wafers
- Good Thermal Conductivity



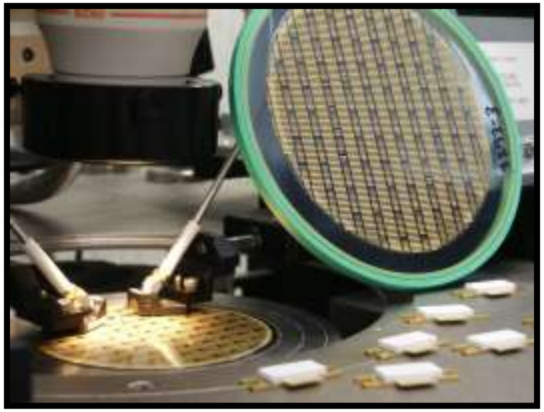
## > GaN-on-Si Epitaxy

- High Quality Crystal Growth
- Proprietary Transition Layers
- Low Screw Dislocation Densities
- No Observed Reverse Piezoelectric Cracking



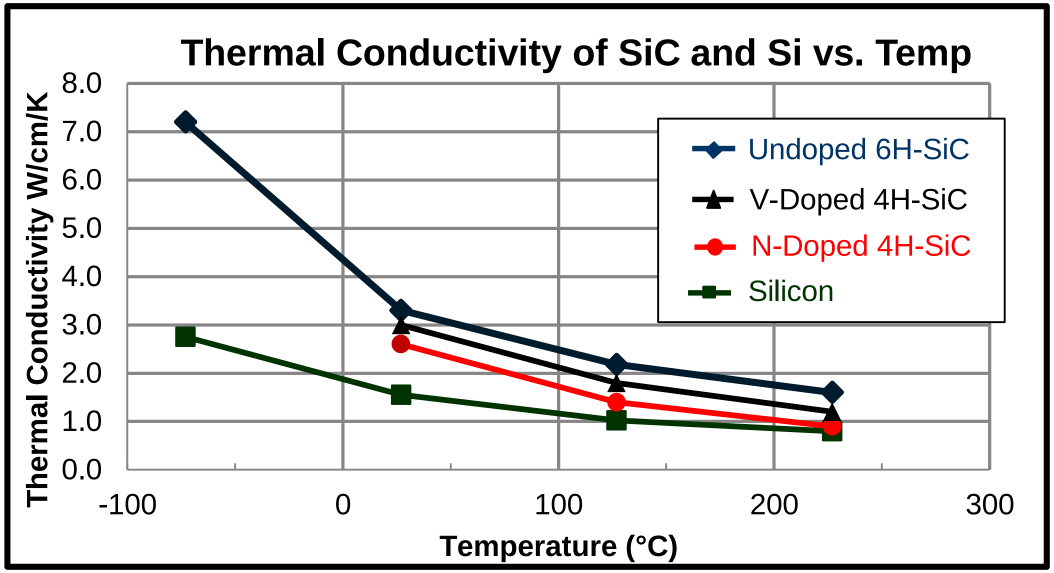
## > Wafer Processing

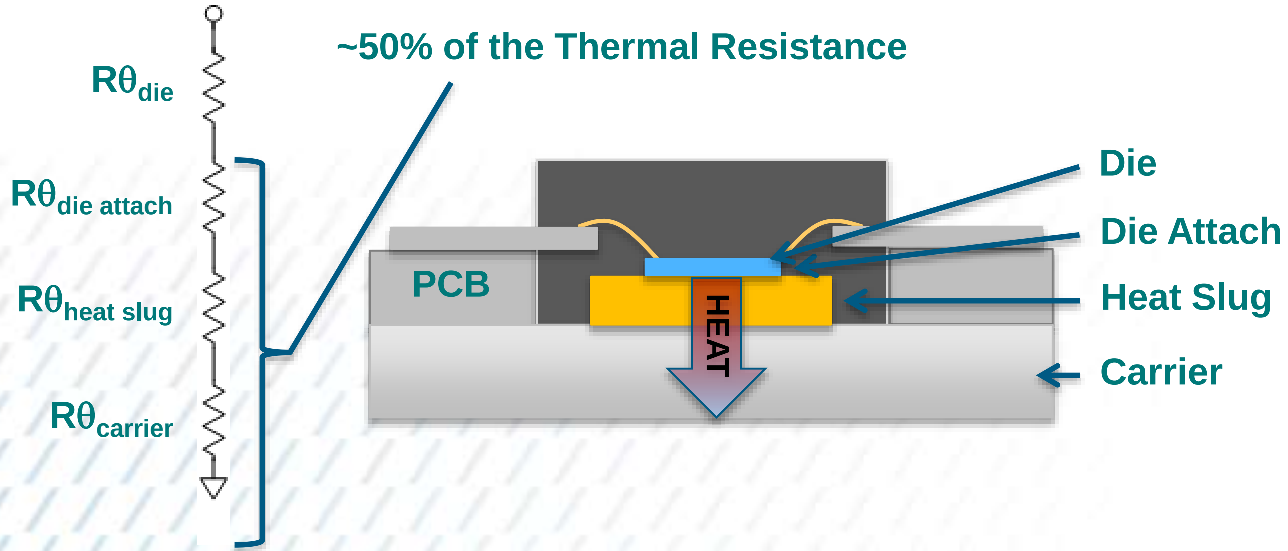
- Ease of Backside Processing
- Leverages Mature LDMOS-like Thinning & VIAs
- Scalable to 200-mm and beyond
- Scalable into CMOS high-volume foundries



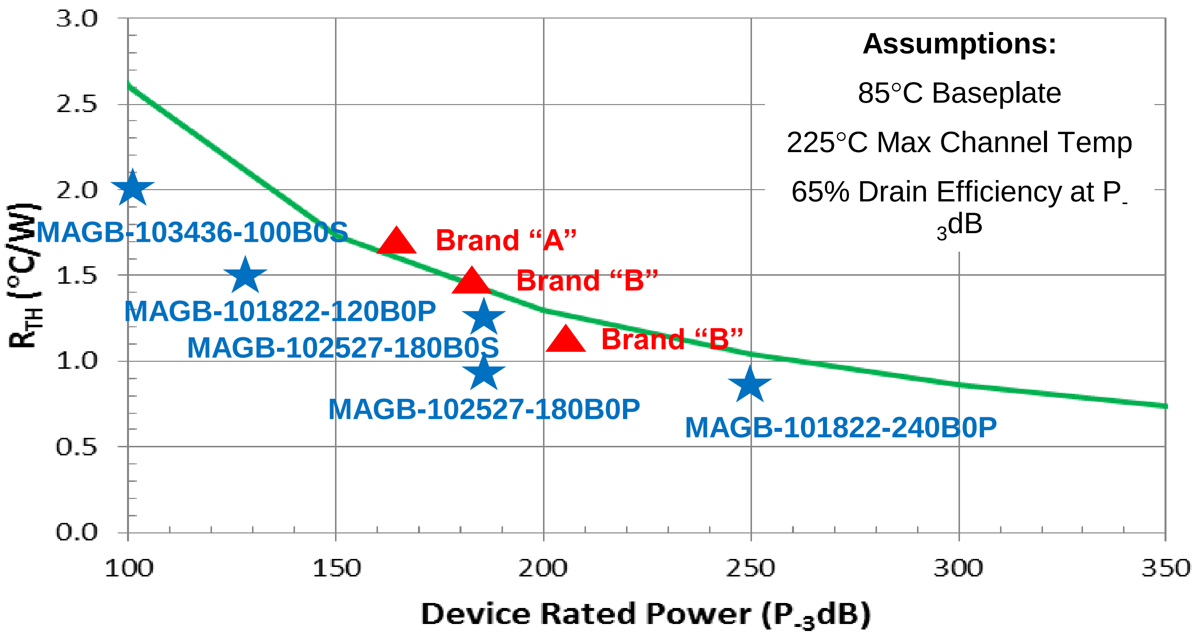
# Is SiC Thermal Conductivity Better?
# Material vs. Product Comparison

> **Product Level Impact**
- Die Thermal Conductivity favors SiC
- As a system, what does this mean?



# Discrete Transistor Thermal Comparisons




| Device | Substrate | Power (Watts) 50 Ohms | Theta J-C |
|---|---|---|---|
| NPT2022 | Si | 100 | 1.3 |
| Brand "A" | SiC | 120 | 1.5 |
| Brand "B" | SiC | 90 | 1.5 |
| Brand "C" | SiC | 100 | 1.0 |
| Brand "D" | SiC | 90 | 2.1 |

- **"Silicon isn't as good a thermal conductor as SiC"**
- **Thermal resistance is function of:**
  - Substrate thickness (Si can be thinned to 2 mils)
  - Power and Voltage (die size)
  - Die Layout
  - Substrate Properties
  - Die Attach
  - Package

**Takeaway:**

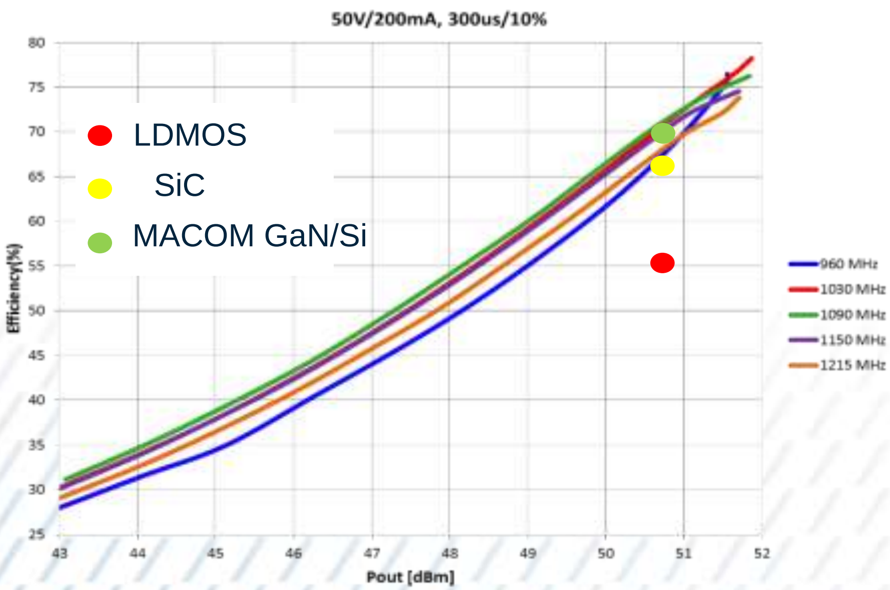
GaN/Si products have similar or even better thermal ratings than equivalent GaN/SiC products

# L and S Band Product Comparisons

## MAGX-100914-125

AC-400S-2

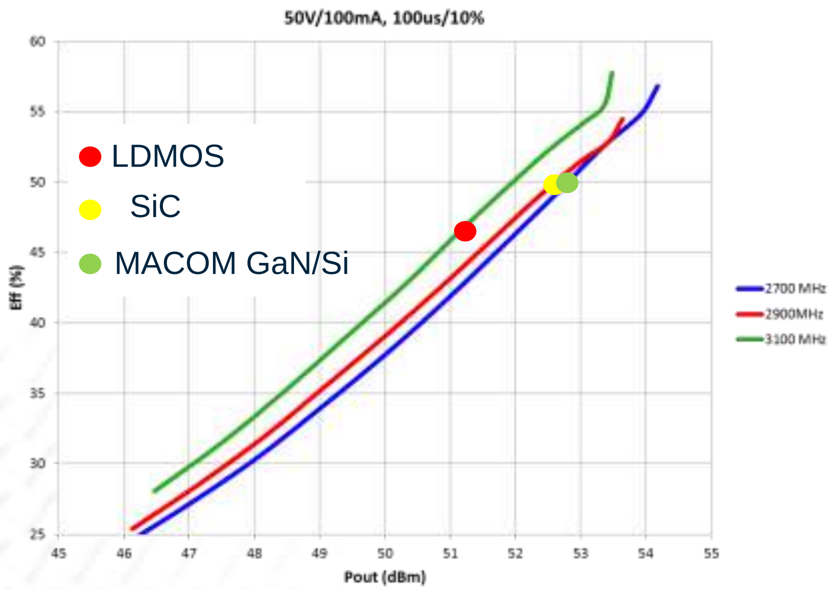
- $V_{DS}$ = 50 V
- Freq. = 960 – 1215 MHz
- $P_{OUT}$ = 125 W
- Pulse Conditions: 300 µs / 10%
- Gain > 17.5 dB
- Drain Efficiency >70%




## MAGX-102731-180

AC-360B

- $V_{DS}$ = 50 V
- Freq. = 2700 – 3100 MHz
- $P_{OUT}$ = 180 W
- Pulse Conditions: 100 µs / 10%
- Power Gain > 14.5 dB
- Drain Efficiency >50%




**GaN/SiC Performance at LDMOS Cost Structures**

# GaN/Si vs LDMOS - 2.45GHz Performance at 2.5dB Compression vs Temperature

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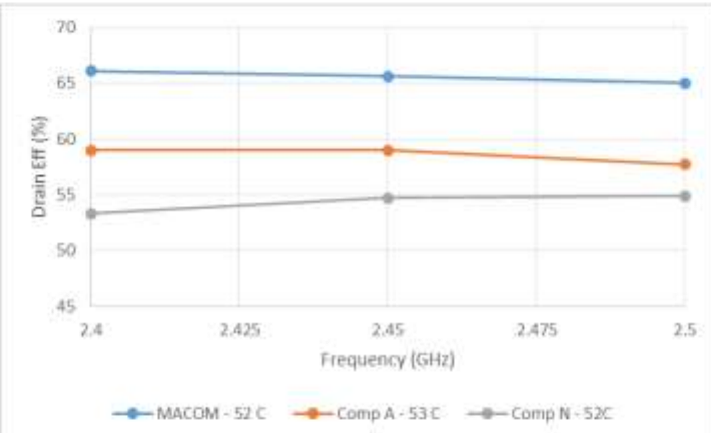
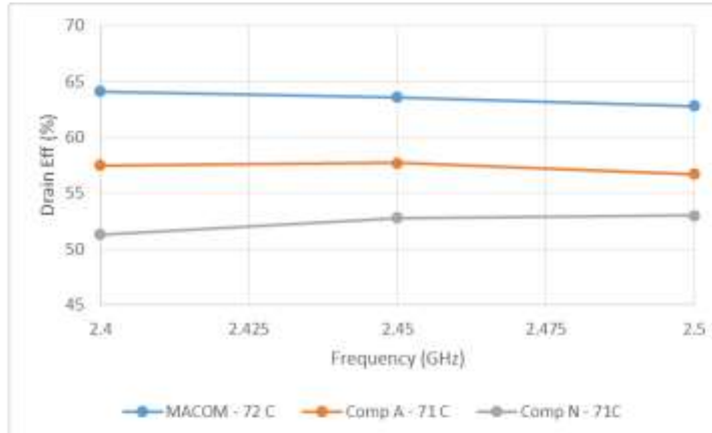
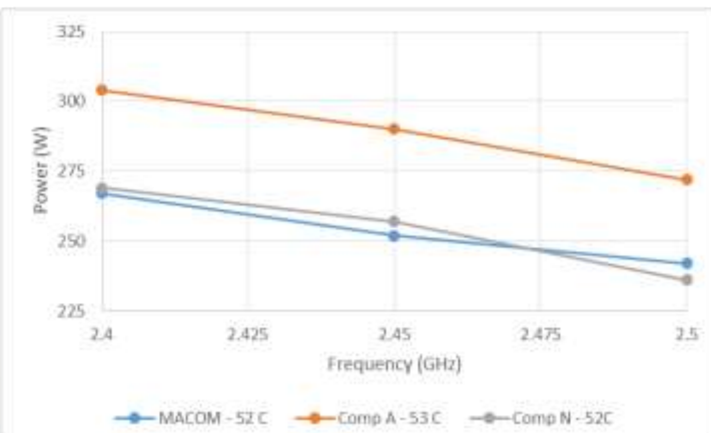
Tc=52 C

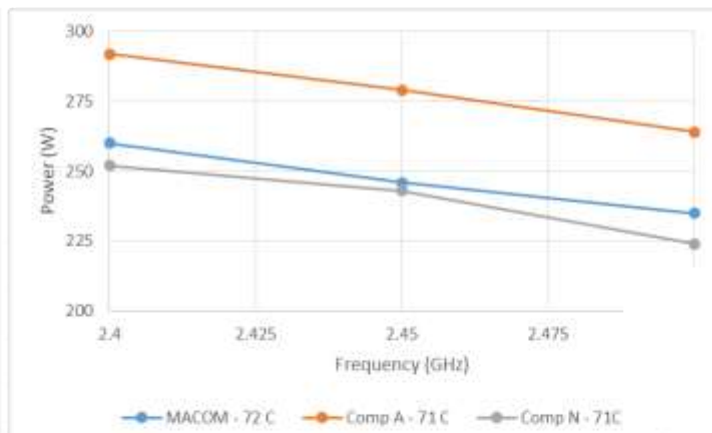
Tc=71 C



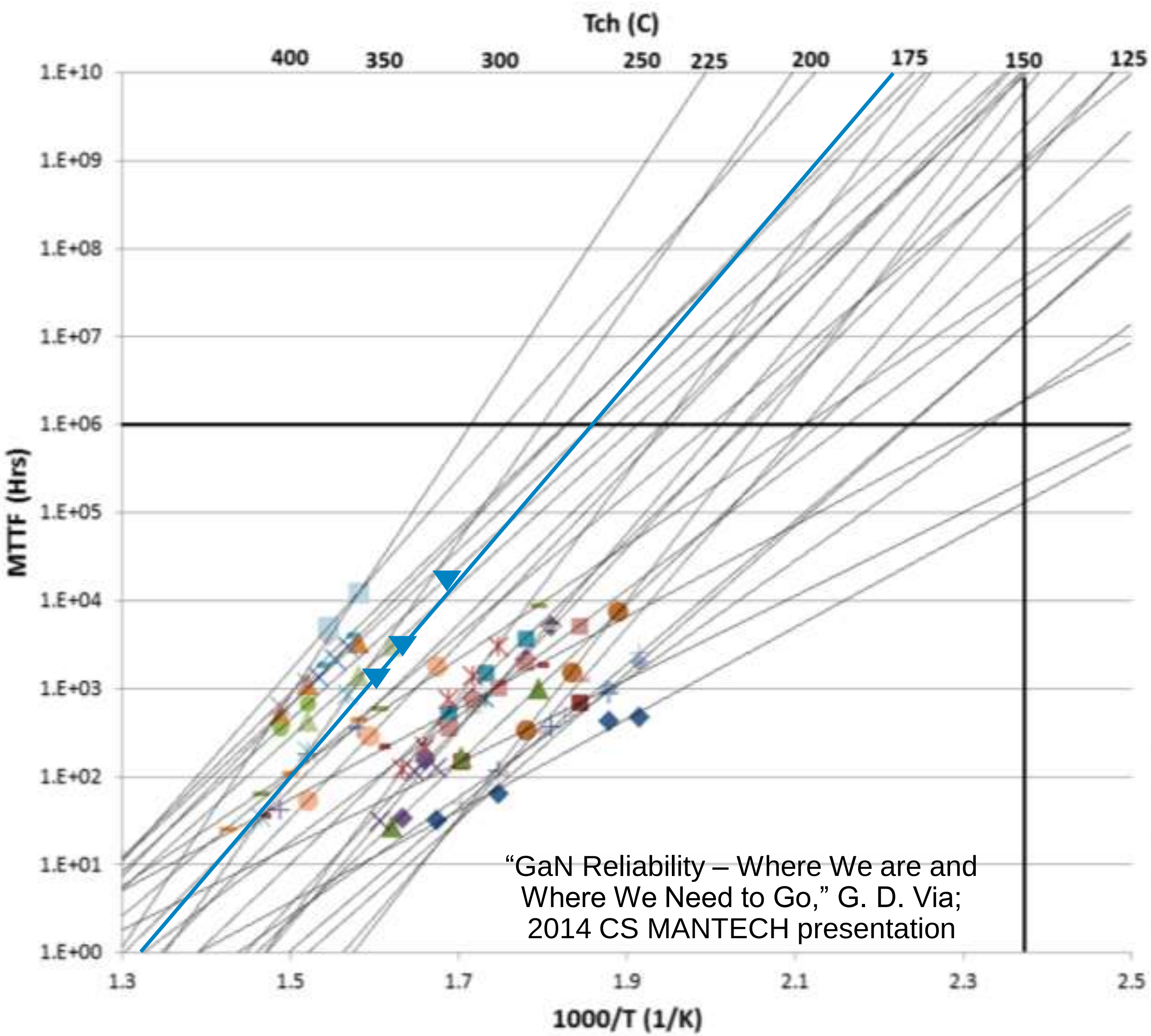

**Drain Efficiency**




**Output Power**

# Reliability -Published GaN Arrhenius Results with Gen4 GaN Overlay




- AlGaN/GaN HEMT on SiC
- Foundry level assessments
- (29) ALT data sets analyzed
- dc and RF tests
- Vd from 20 to 65 V
- $T_{ch}$ from 250 to over 400°C
- Test times from $10^1$ to $10^{4*}$ hrs
- $E_a$ from 1.0 to 2.5 eV
- MTTF @ 150 C > $10^6$ Hrs

**Gen4 GAN:**

- MTTF @ 225 C =3.1x $10^6$ Hrs
- $E_a$ = 2.1 eV

# GaN-on-Si Enables Homogeneous Integration With Silicon Devices

- Integration Roadmaps for GaN-on-Silicon include homogeneous integration of GaN devices and Si-based devices on the same chip.
- **Si-CMOS and GaN-on-Si device integration has already been demonstrated** by third party development efforts.

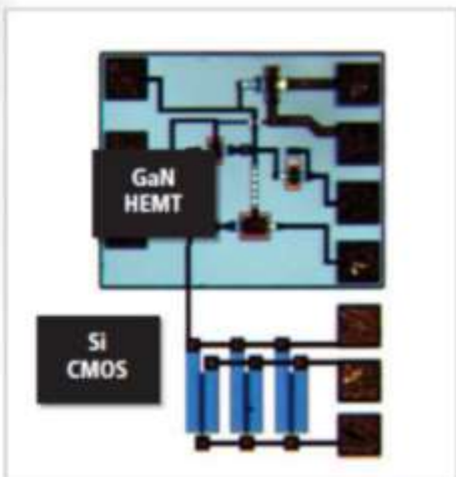
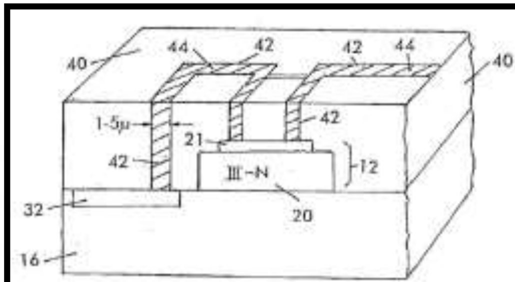



*Figure 2. Optical micrograph of the first GaN and Si CMOS heterogeneously integrated MMIC.*

- The demonstrated circuit to the left serves as a **building block for digitally assisted RF and mixed signal circuits**, such as amplifiers with on-chip digital control and calibration, reconfigurable or linearized PAs with in-situ adaptive bias control, high-power digital-to-analog converters (DACs) and on-chip power distribution networks.

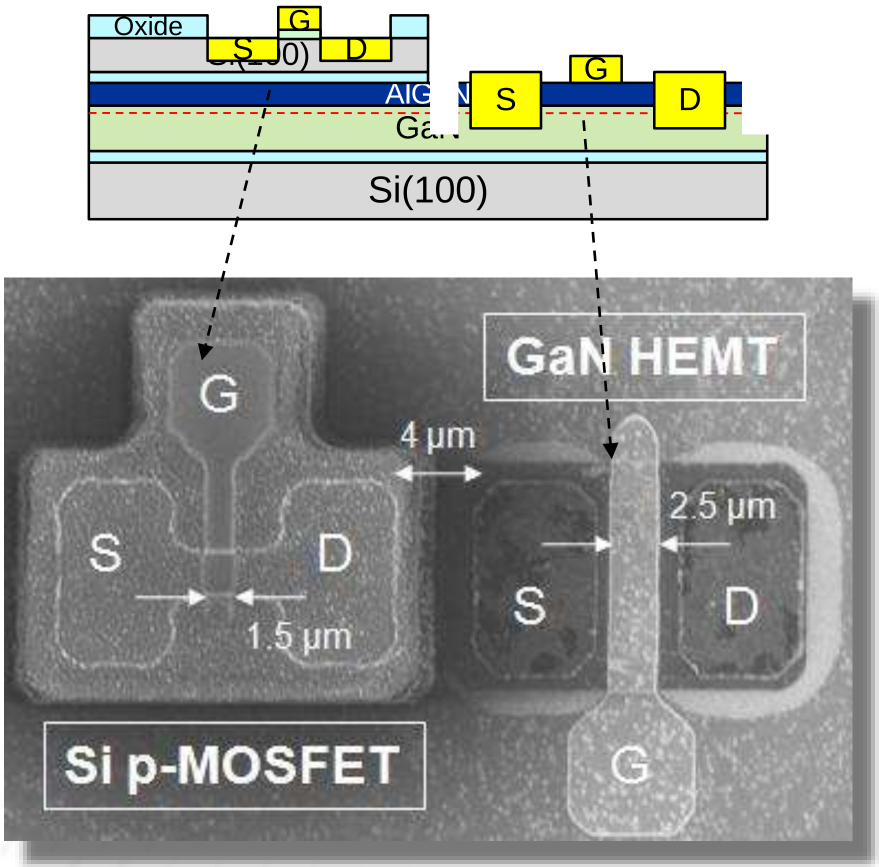
**GaN-on-SiC is incompatible with homogeneous integration with Si-devices on a common chip**



# GaN Future Directions

## > GaN/MOSFET Integration

- Integration of III-V HEMTs and Si (100) MOSFETs
- High power digital-to-analog converters (DACs)
- On-wafer wireless transmitters
- Driver stages for on-wafer optoelectronics
- Power amplifiers coupled to Si linearizer circuits
- High speed (high power) differential amplifiers
- Normally-off power transistors
- New enhancement-mode power transistors
- Buffer stages for ultra-low-power electronics



# Conclusions

Silicon is a more mature and defect-free substrate for GaN devices and meets the volume ramp requirements for BTS and RFE market adoption

GaN/Si products deliver similar thermal performance to GaN/SiC

GaN/Si A&D devices have demonstrated equal or even superior performance to similar GaN/SiC devices

GaN/Si RFE devices have demonstrated superior efficiency and efficiency flatness to LDMOS devices

GaN/Si uniquely enables homogeneous integration

**GaN/Si is ideally suited to service volume RF markets like Basestations and RF Energy**